Atharv Ramesh Nair

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RESEARCH INTERESTS : LLM Inference, AI Safety, Interpretability, Reinforcement Learning, Medical Imaging.

Education

University of California San Diego La Jo

M.S. in Machine Learning and Data Science (ECE) ***GPA: 4.0/4.0*** *Sep 2025 -*

Coursework: Reinforcement Learning, Deep Generative Models, Efficient AI, ML Systems, Statistical Learning, Linear Algebra

Indian Institute of Technology Hyderabad (IITH) Hyderabad, India

B.Tech. in Electrical Engineering ***GPA: 3.91/4*** *Nov 2020 - May 2024*

Coursework: ML, DL, NLP, CV, Probability, Information Theory, Algorithms, Convex Optimization

Research & Teaching Experience

Trustworthy ML Lab (Prof. Lily Weng) — Spatial Concept Bottleneck Models UCSD, Nov 2025 - Present

- Developing **spatially interpretable concept bottleneck models** that expose where each concept is grounded in an image, enabling prediction audits at both concept and region level.
- Built a medical-imaging CBM testbed on **CheXpert** (**LF-CBM, SALF-CBM, VLG-CBM**), evaluating concept fidelity and localization with VLM judges and Grad-CAM++ to identify grounding failures.

Teaching Assistant – DSC 291: Trustworthy ML UCSD, Spring 2026

- Delivered a lecture on **LLM jailbreaks and defenses** covering attack methods, mitigation strategies, and live demos; also present hands-on demos across trustworthy ML topics.

Deep Learning for OCT Imaging (Dr. Kiran Vupparaboina) University of Pittsburgh, Jan 2024 - Jun 2024

- Co-authored a **Bioengineering 2024** paper: fine-tuned RETFound on **1,770** OCT B-scans, beating single-task/multi-task/ResNet-50 baselines at **0.75-0.80 AUC-ROC**; built a MONAI generation pipeline.

Publications

Du, K.; **Nair, A.R.**; et al. *Detection of Disease Features on Retinal OCT Scans Using RETFound*. *Bioengineering*, 2024, 11, 1186. https://doi.org/10.3390/bioengineering11121186

Select Projects

DFlash: Accelerating LLM Inference with Speculative Decoding

- Extended **DFlash**, a speculative decoding system with a **block diffusion draft model**, across Transformers (**FlashAttention**) and SGLang (**FlashInfer**) with multi-candidate verification and reproducible benchmarks.
- Best AIME25 run improved throughput 129.25→146.60 tok/s (**+13.4%**) and mean accepted length 6.43→7.58, showing gains depend on verification cost and backend execution constraints.

Prompt-Length Optimization via Reinforcement Learning

- Cast adversarial suffix optimization as sequential decision-making: an RL agent chooses token add/delete/keep edits, and each candidate is refined with **Greedy Coordinate Gradient** search.
- Benchmarked **GRPO**, PPO, and REINFORCE for stability and sample-efficiency; the final system cut suffix length **43.8%** (16→9 tokens) while improving attack log-likelihood from **-1.64→-1.12**.

StealthRL (under review at ACL ARR)

- Built an RL-based paraphrase attack with **GRPO + LoRA** on Qwen3-4B, optimizing multi-detector evasion plus E5 semantic similarity to stress-test AI-text detectors.
- Showed catastrophic failure at security-relevant settings: mean TPR@1%FPR fell to 0.024, mean AUROC dropped 0.79→0.43, and attack success reached 97.6%, transferring to held-out Binoculars and MAGE detectors.

Safety-Aware Speculative Decoding

- Extended Speculative Safety-Aware Decoding with a Composite Risk Score over draft-target disagreement, cutting attack success 12%→4% while halving over-refusal versus vanilla SSD.
- Added training-free prompt and perplexity safeguards plus a 4-bit pipeline; on 105 prompts, achieved a 67% relative ASR reduction with 10% benign over-refusal.

Industry Experience

Netradyne – *Software Engineer — Machine Learning (Edge / Perception)* Bengaluru, India - *Jun 2024 - Aug 2025*

- **Real-time edge inference:** Built multi-camera, multi-model perception pipelines for in-cabin and road-facing tasks on embedded accelerators and GPUs, including driver monitoring and collision alerts.
- **Inference stack re-architecture (-17% latency):** Reworked the video system with producer-consumer scheduling, async I/O, and priority queues, eliminating frame drops under multi-model load.
- **Serving reliability & deployment:** Consolidated services into one multithreaded daemon (**-3% RAM**), debugged Qualcomm memory/concurrency bottlenecks, and standardized ONNX builds and Docker CI.

Awards

1st Place — IEEE Signal Processing Cup 2024 (ICASSP) — Far-Field Speaker Verification; **2nd Runner-Up** — IEEE Video and Image Processing Cup 2023 (ICIP) — OCT Biomarker Detection.

Skills

ML / Inference PyTorch, Transformers, vLLM, SGLang, FlashAttention, FlashInfer, LoRA, ONNX, TensorRT
Programming Python, C/C++, MATLAB, SQL, Bash
Systems / Infra Linux, Docker, Kubernetes, Git, CI/CD, AWS (EC2, S3)
Data / Tooling NumPy, Pandas, SciPy, Matplotlib, PostgreSQL, MongoDB